



Cost Foresight: Forecasting Steel Production Costs



THE BUSINESS CASE FOR COST FORECASTING

Why Predicting Cost per Ton Is Critical for Strategic and Operational Decisions

Industry Cost Exposure

Steel producers face extreme volatility driven by:

- Scrap metal price shocks
- Energy and fuel markets
- Shipping and logistics
- Global disruptions (COVID-19, Ukraine War)

Managerial Need

Forecasting cost per ton helps managers:

- Protect margins
- Set competitive pricing
- Plan budgets under uncertainty
- Allocate resources efficiently



Integrating Financial, Operational, and Macroeconomic Signals

Data Foundations of the Project



Key Sectors

Internal Financial Data

(From SEC Filings & Annual Reports)

- Cost of Products Sold (COGS)
- Steel shipments
- Inventory levels
- Scrap usage
- These form the operational backbone of cost behavior.

External Market Drivers

(From BLS, EIA, Census, NOAA)

- Scrap price (PPI)
- Electricity & natural gas
- Diesel & rail freight
- Export volumes
- These capture global supply-demand and input pressures.

Shock & Risk Indicators

(COVID-19, Ukraine War, Macroeconomic stress)

- NOAA disaster counts
- Policy uncertainty index
- Market volatility measures
- These explain “structural breaks” in cost patterns.

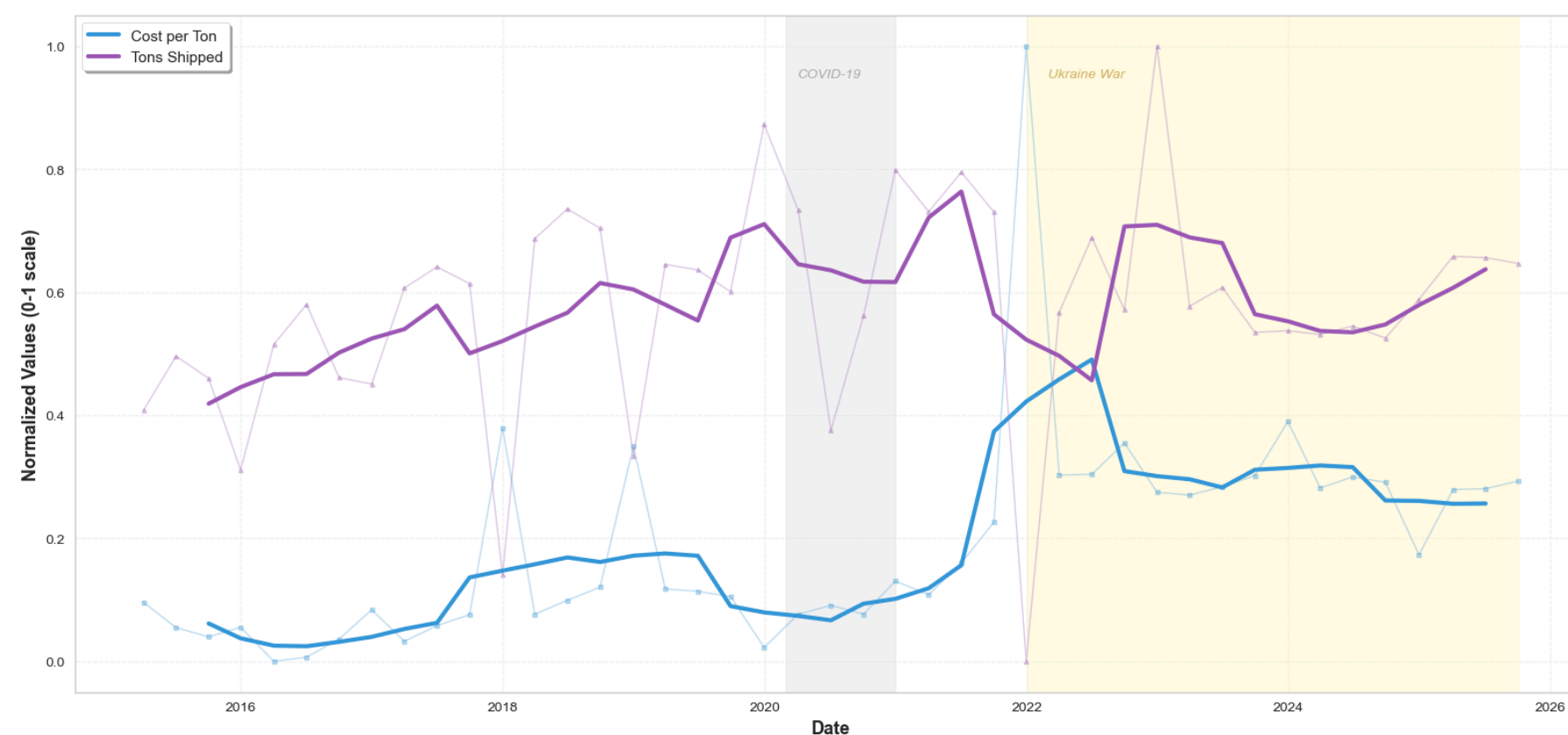
Efficiency & Productivity Metrics

(Internally engineered)

- Inventory turnover
- Production volume dynamics
- Rolling volatility features
- These reveal operational performance and cost discipline.

KEY TRENDS & COST DYNAMICS

What the Data Reveals About Cost Behavior Over Time



Treand analysys - Cost per ton vs Tons shipped (normalized)

Extreme Cost Volatility

Cost per ton shows violent spikes:

- Sharp increases during the Ukraine War energy crisis
- COVID-19 supply chain shocks
- These periods break historical patterns completely.

Economies of Scale

Higher steel shipments → lower cost per ton (negative correlation visible across most years).

Crisis-Driven Anomalies

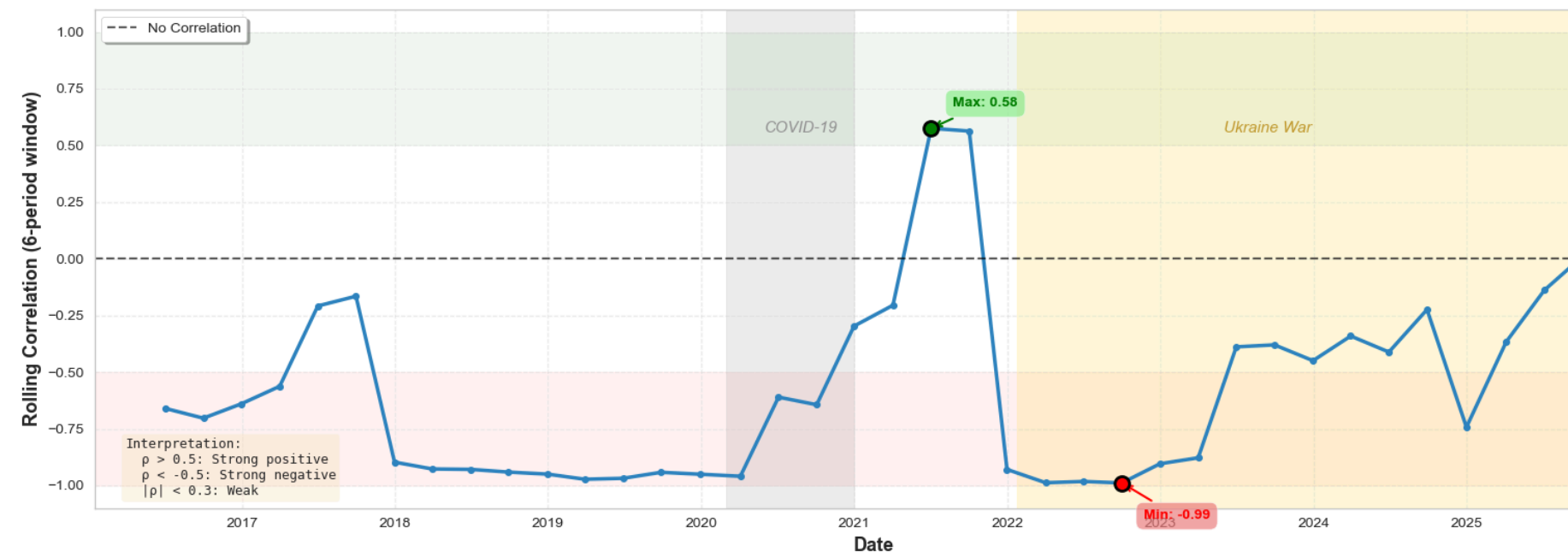
During 2021–2022, volumes collapsed while costs exploded, reversing the usual cost-volume relationship.

Structural Breaks

Z-score analysis shows 2022 Q2 cost levels were 4.8 standard deviations above normal basically a statistical “black swan.”

Feature Engineering: Turning Data Into Signals

Building Meaningful Predictors for Cost Forecasting



Rolling Correlation (6-quarter window) between Cost per Ton and Tons Shipped.

Rolling Correlation Signals

Cost-volume correlation flips across periods (COVID, Ukraine War), so rolling features capture regime changes.

Lagged Variables for Market Delays

Many cost drivers (e.g., scrap exports) impact Nucor one quarter later, so lagged features capture delayed effects.

Volatility Indicators

3-quarter rolling volatility detects unstable cost environments and shock periods.

Autoregressive Cost Behavior

Past cost levels (Y_{t-1} , rolling Y) stabilize predictions and capture pricing inertia.

Predictive Modeling Strategy

Building Robust Forecasts Under Limited and Volatile Data

Cost Forecasting Methodology

Regularized Linear Models

Ridge, Lasso, Elastic Net used to handle multicollinearity across energy and commodity variables.

Time-Series Cross Validation

Used TimeSeriesSplit to respect temporal order and avoid look-ahead bias.

Baseline Benchmark

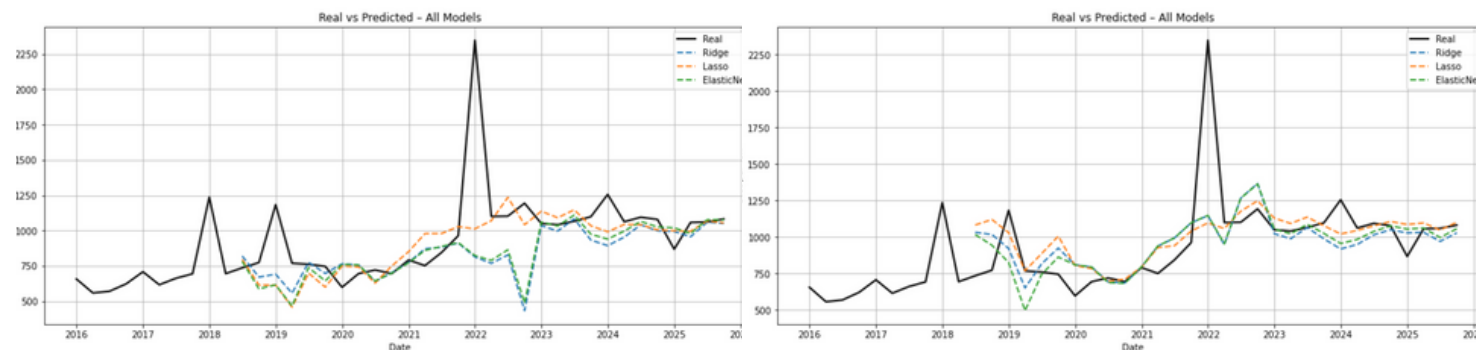
A naive “next quarter = this quarter” forecast sets the minimum performance threshold.

Deep Learning Feasibility Test

LSTM & GRU models overfit heavily due to only 43 observations, confirming simpler models are more reliable.

Forecasting Model Results

How Well the Models Predict Cost per Ton

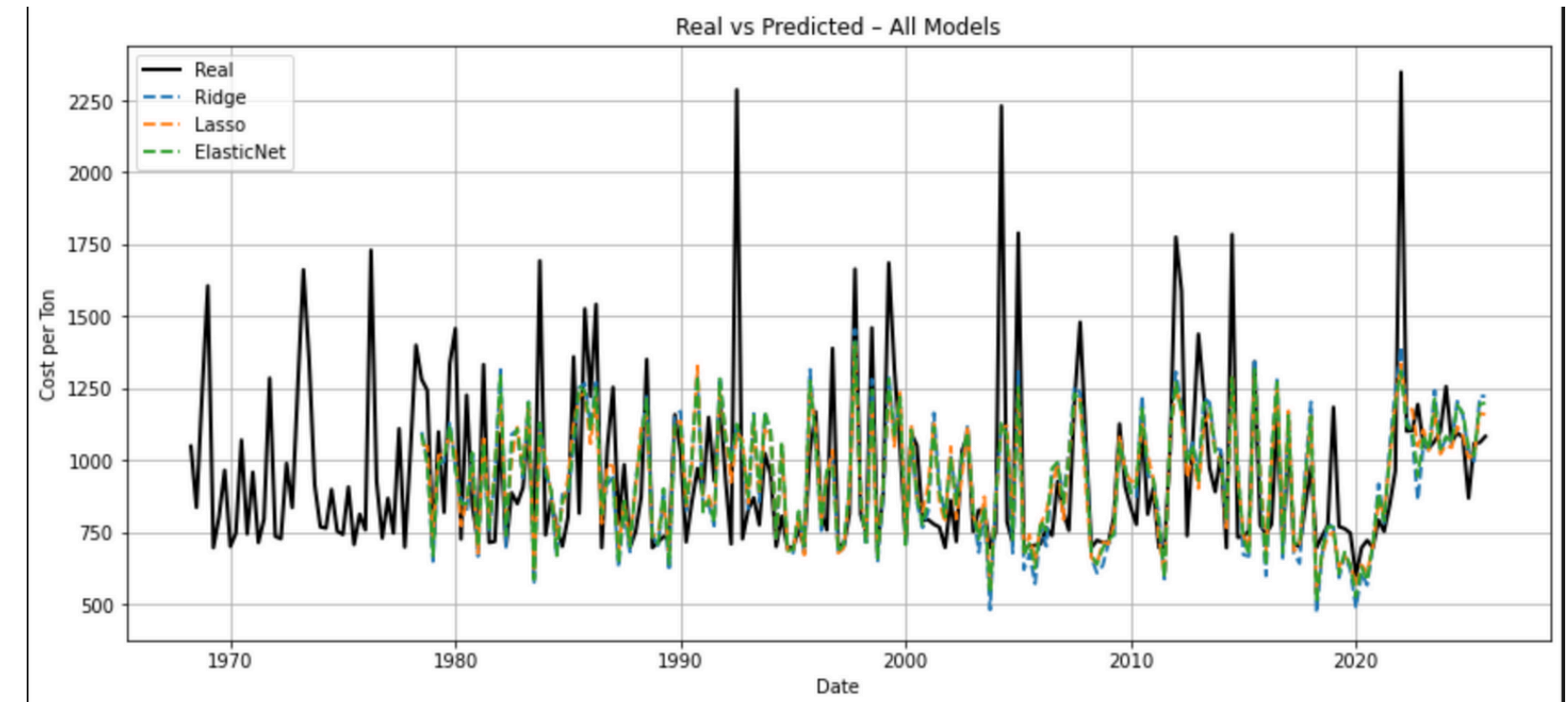


Lasso Delivers the Most Reliable Forecasts Under Limited Data

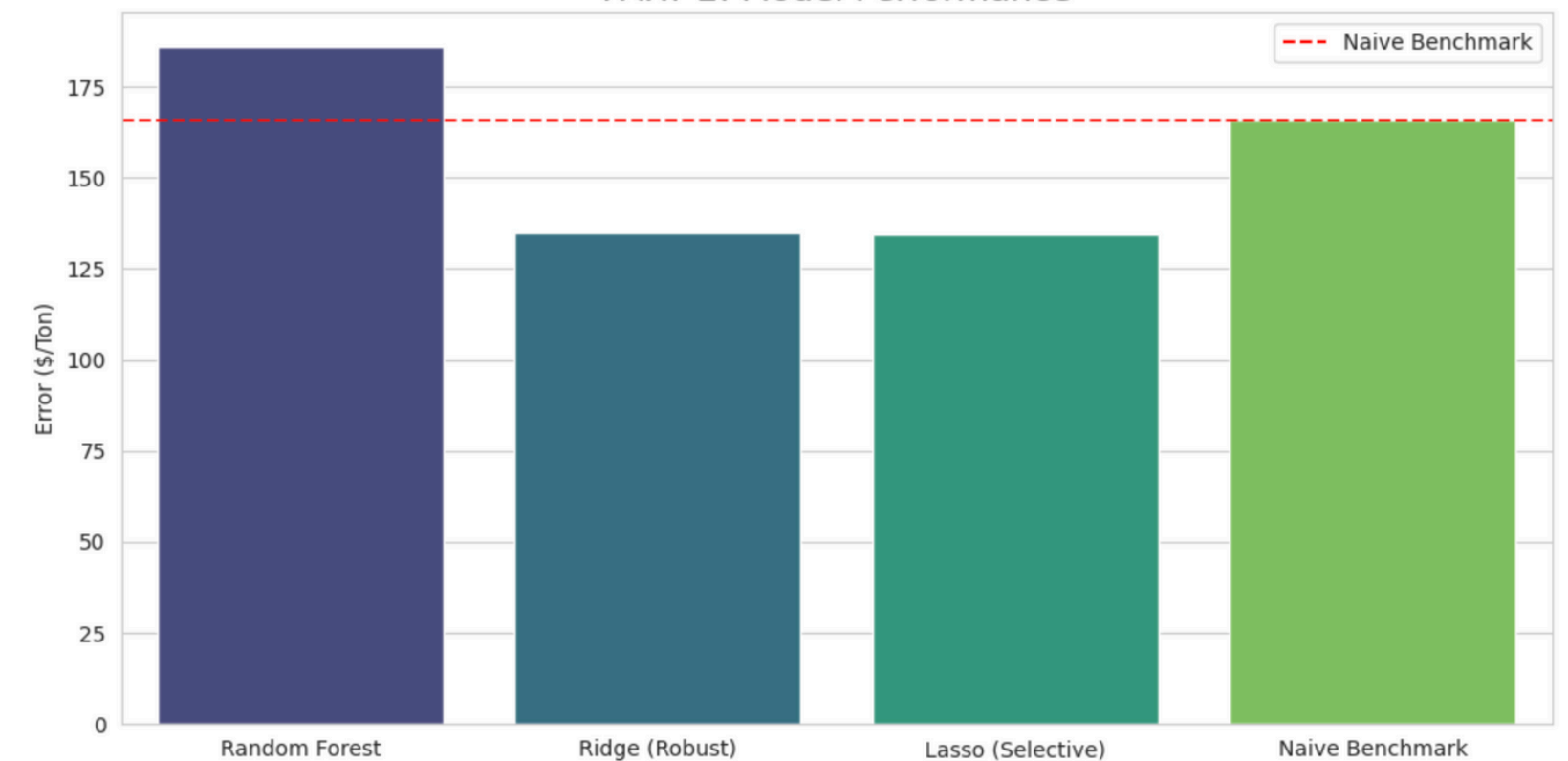
Despite only 43 quarters of real data and high market volatility, Lasso remains the most stable model. It outperforms Ridge, Elastic Net, and even the naive benchmark by controlling overfitting and capturing the key cost drivers more effectively.

Lagged Features and Synthetic Augmentation Improve Performance in Volatile Periods

Dynamic inputs (like lagged scrap exports and Y_{t-1}) help the model adapt during shock periods such as 2021–2022. When synthetic copula data is added, the model becomes more consistent and achieves positive predictive accuracy — showing that data scarcity, not model design, is the real limitation.

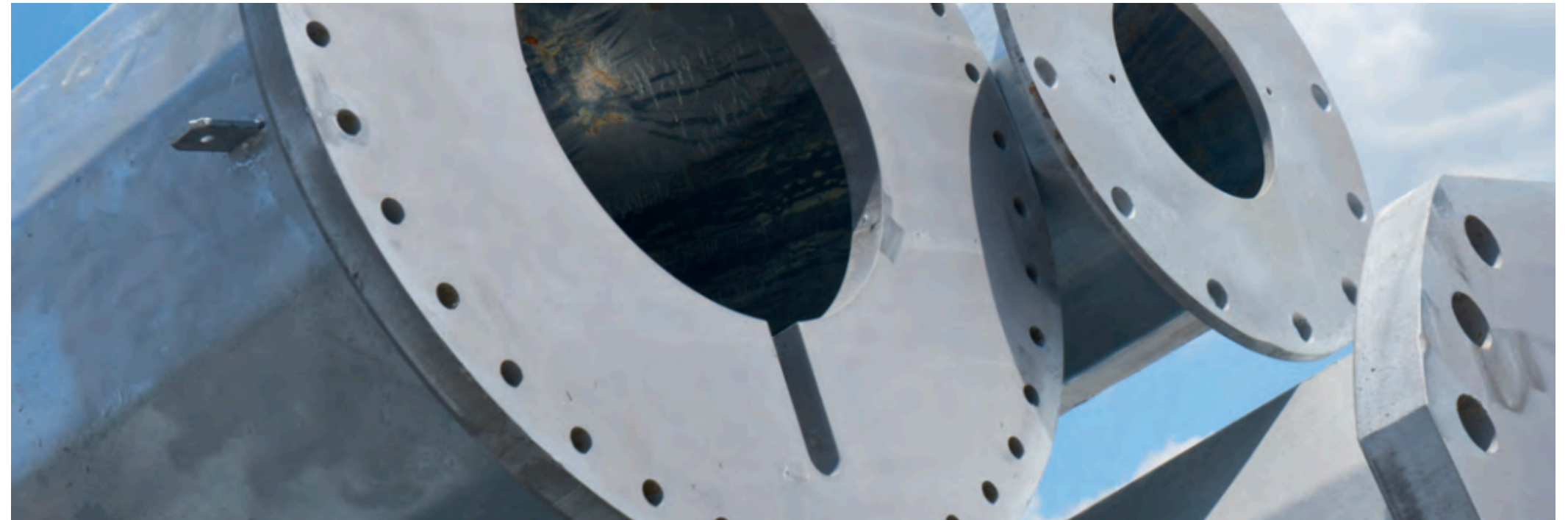


PART 2: Model Performance



Managerial Insights

What the Forecasts Mean for Strategic & Operational Decisions



Cost Planning & Financial Stability

- Forecasts help build buffers for high-cost periods.
- Improve budget accuracy and reduce unexpected margin erosion.

Pricing & Operational Decisions

- Supports better pricing strategies during volatile markets.
- Helps optimize production timing, energy usage, and inventory management.

Risk Management & Resilience

- Early detection of cost shocks strengthens crisis readiness.
- Aligns with Nucor's strategic focus on resilience and long-term sustainability.
- (Directly supported by Nucor's annual report risk section.)

Limitations & Future Opportunities

Improving Forecast Accuracy Through Better Data and Broader Market Signals



Data Constraints

- Only 43 quarterly observations → limits model learning.
- Low-frequency data misses month-to-month cost behavior.
- Structural breaks (COVID, Ukraine War) reduce model stability.



Modeling Enhancements

- Explore hybrid models (ARIMAX, Prophet + regressors).
- Add seasonality and demand-side indicators.
- Build scenario-based simulations for crisis conditions.



Expanded Market Signals

- Integrate global steel demand trends.
- Include futures prices, energy benchmarks, and supply chain indicators.
- Track policy, tariff, and geopolitical risk more explicitly.

Business Benefits of Cost Forecasting

What the Forecasts Mean for Strategic & Operational Decisions



Understanding of Cost Drivers

- Integration of internal financial metrics with macroeconomic indicators
- Clarifies key drivers of cost per ton
- Highlights impact of scrap prices, energy costs, logistics, and inventory dynamics

Insights into Temporal Effects

- Lagged features offer marginal gains during structural shocks
- Shows importance of capturing short-term momentum
- Highlights inventory-driven dynamics

Foundation for Future Modelling Improvements

- Establishes a solid methodological base for advanced models
- Includes a validated feature framework
- Clarifies which modelling strategies work under different economic conditions
- Provides a strong starting point for future research



CONCLUSION

Insights That Strengthen Financial, Operational, and Strategic Decision-Making

Simpler models work best in volatile, small-data environments

Deep learning and non-linear models overfit heavily, while Lasso/Ridge consistently produced the strongest and most reliable out-of-sample results.

Synthetic data is essential but not perfect

Copula augmentation stabilised the models and raised R^2 from negative to positive, but it cannot replicate real crisis shocks. It is useful for scenario analysis, not precise quarter-ahead prediction.

Regime-Dependent Value of Lagged Features

Lagged features are not consistently beneficial. Outside high-volatility regimes, costs are driven mainly by inflation, energy prices, and scrap markets.

The framework still holds strong strategic value

Even with limits, the model supports Nucor by helping anticipate cost uptrends, adjust prices, plan scrap contracts, and manage production with more confidence.

Thank you for the attention!

Questions & Discussion